
BrainDyn: A Sheaf Neural ODE for Generative Brain Dynamics

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Abstract

Efficient neural network models that generate brain-like dynamic activity can be a valuable resource for generating synthetic data, analyzing differences in brain transients under conditions such as testing perturbation activity or inferring the underlying generative dynamics. However, large language models (LLMs) or standard recurrent neural networks (RNNs) ignore the anatomical organization and therefore do not produce components that align with brain regions. On the other hand, graph-based networks often have very simple message passing rules that are not sufficiently expressive for brain-like dynamics. To address this, we introduce BrainDyn, a sheaf neural ordinary differential equation (neural ODE) model for continuous-time dynamics on structured brain graphs. BrainDyn encodes the recent activity history of each brain region using a long short-term memory (LSTM) model over a sliding temporal window to produce hidden states, or *stalks*, that are projected through learnable restriction maps into edge-specific shared spaces. Discrepancies between neighboring nodes in these shared spaces are characterized by a sheaf Laplacian that can facilitate message passing between neuronal units. The output of these messages is then fed to a neural ODE that governs the continuous-time evolution of neuronal activity. We evaluated BrainDyn on resting-state fMRI (PNC dataset), scalp EEG with focal epilepsy (TUSZ dataset), and simulated activity from the NEST spiking network simulator. BrainDyn achieves strong forecasting ability across modalities, and the resulting representations support downstream tasks including *in silico* perturbation prediction.

1 Introduction

Generative AI has begun to serve as a virtual modeling tool for complex biological systems, including cells, tissues, and patients [1–3] at various levels of abstraction. Building an analogous virtual model of the brain remains more difficult [4, 5]. Brain activity is transient, coordinated fluctuations over a large population of neural units that unfold over time [6, 7]. These dynamics are central to cognition, behavior, and disease, yet many commonly used descriptors such as mean activation maps or pairwise functional connectivity reduce neural activity to static summaries. Such summaries are useful, but they discard the temporal structure needed to model how brain states emerge, propagate, and change [8–11]. A generative virtual brain model has to synthesize brain-like transients that

faithfully reflect the spatiotemporal structure of real neural activity, and produce representations that support downstream applications such as perturbation testing [12].

Efficient generative models of neural dynamics are therefore a practical resource: they can produce synthetic data to augment limited recordings, model differences in brain transients across conditions, and serve as *in silico* testbeds for perturbation analysis [13, 14]. However, existing architectures fall short in distinct ways. Standard sequence models, including recurrent networks and transformers, can capture temporal dependencies but are agnostic to anatomical organization; their predictions cannot be attributed to specific brain regions or constrained by the network structure that shapes real neural communication. Graph neural networks offer a more natural fit, with parcellated regions, EEG channels, or individual neurons as nodes and anatomical or functional relationships as edges [15, 16]. However, standard message passing architectures carry their own critical limitation: all nodes communicate in the same feature space, with aggregation performed by scalar summation or averaging [17, 18]. In the brain, this assumption is questionable, since distinct regions differ systematically in the geometry and content of their representations [19, 20]. Repeated scalar aggregation leads to oversmoothing, progressively homogenizing regional activity and erasing the heterogeneous dynamics that are a hallmark of neural computation [21, 22].

A principled remedy is the cellular sheaf, a mathematical structure that equips each edge with linear maps, called restriction maps [23–25], that transform node features into an edge-specific shared space before aggregation. Sheaf neural networks [26–28] generalize standard GNNs by allowing each connection to implement a structured, feature-wise transformation, enabling brain regions to maintain distinct representational geometries while communicating in a coherent framework. This expressivity is not merely a modeling convenience; it reflects a biologically motivated hypothesis that communication between brain regions involves transformation, not just summation.

We introduce **BrainDyn**, a sheaf neural ordinary differential equation (neural ODE) model for generative brain dynamics. BrainDyn associates each node and edge with a vector space and learns restriction maps that embed regional activity into edge-specific shared spaces. Temporal encoders summarize the history of activity at each node, and an attention mechanism [29] modulates restriction maps to capture how each connection dynamically shapes information flow. The resulting sheaf Laplacian, derived from these maps and encoding the discrepancy between transformed signals, is embedded within a neural ODE [30–32] that continuously evolves neural activity forward in time. This architecture enables both forecasting and generation of brain-like transients, supports *in silico* perturbation, and yields structured latent representations suitable for downstream analysis.

We evaluate BrainDyn on three neural recording modalities: the Philadelphia Neurodevelopmental Cohort (PNC) resting-state fMRI corpus [33], the Temple University Hospital Seizure (TUSZ) EEG corpus [34], and simulated activity from the NEST spiking neural network simulator [35]. On NEST simulations, where ground truth generative structure is known, we validate *in silico* perturbation capabilities, showing that BrainDyn accurately models and responds to controlled interventions. Together, these results establish BrainDyn as a step toward a principled and expressive virtual brain.

To summarize, our main contributions are:

1. **Sheaf-based neural dynamical model:** We introduce **BrainDyn**, a sheaf-based neural ODE framework that integrates sheaf-based message passing to model continuous-time neural dynamics on structured brain networks.
2. **Expressive interaction modeling:** BrainDyn represents interactions between brain regions with learnable restriction maps, enabling structured, feature-wise transformations that captures complex, heterogeneous interactions across connections.
3. **Generative modeling of neural dynamics:** BrainDyn generates brain-like trajectories that capture neural transients and temporal evolution, and reflect the underlying interactions across brain regions.
4. **Comprehensive evaluation across modalities and tasks:** We evaluate BrainDyn across multiple neural data modalities, including fMRI, EEG, and NEST simulations, achieving strong performance in modeling neural dynamics, supporting downstream tasks, and capturing controlled perturbations.

2 Related Works

Sequential modeling of brain dynamics. Neural population activity is often organized by low-dimensional dynamics: coordinated trajectories through state space that support computation, behavior, and brain-state transitions [36–38]. Recent work has reframed brain signal modeling as a forecasting problem, where the goal is to predict future multivariate activity from recent history [39, 40]. Sequence models provide strong temporal encoders, with some architectures aggregating spatial information through convolutions before feeding the representations into recurrent layers [41, 42], while biosignal transformers such as BIOT [43], model long-range temporal dependencies through attention mechanisms. However, these models are largely agnostic to anatomical or functional organization: brain regions or channels are typically treated as independent, sequential, or fully connected units rather than as nodes in a structured neural system.

Spatiotemporal modeling of brain activity using dynamic graph neural networks. Graph neural networks provide a natural inductive bias for brain data, where parcellated regions, electrodes, or neurons can be represented as nodes and their anatomical or functional relationships as edges. Diffusion-based graph models have been effective for seizure detection and other tasks [39], while dynamic graph architectures such as EvolveGCN allow graph filters to evolve with temporal context [44]. In parallel, neural ordinary differential equations provide a continuous-time framework for modeling latent dynamics [30], including irregularly sampled time series [31] and graph-structured dynamical systems [32]. Recent graph neural ODE models such as ODEBRAIN [45] and RiTINI [46] combine graph message passing with continuous-time evolution, demonstrating the value of relational dynamics for neural and biological systems. However, standard graph and graph ODE models typically assume that all nodes communicate in a shared homogeneous feature space [32, 45]. This assumption is restrictive for brain data, where distinct regions may encode information in different representational geometries, and repeated scalar aggregation can over-smooth regional activity.

Sheaf neural networks and relation to BrainDyn. Cellular sheaves offer a principled way to model heterogeneous interactions on graphs by assigning vector spaces to nodes and edges, together with restriction maps that transform node features into edge-specific shared spaces. The resulting sheaf Laplacian generalizes the graph Laplacian by measuring disagreement only after neighboring node features have been aligned through these maps [26]. Sheaf neural networks extend this construction to learnable graph representation learning, with Neural Sheaf Diffusion showing that nontrivial sheaves can mitigate over-smoothing and handle heterophilic structure more effectively than standard GNNs [28]. Related work with connection Laplacians further links sheaf learning to local geometric alignment [27]. BrainDyn builds on these ideas but targets a different setting: continuous-time neural dynamics. By embedding learnable sheaf-based interactions inside a neural ODE, BrainDyn combines temporal forecasting, structured graph dynamics, and heterogeneous edge-specific transformations, providing a dynamics-grounded framework for modeling neural activity across modalities and perturbation settings.

A more comprehensive discussion of related works can be found in Appendix A.

3 Background

In this section, we will describe the mathematical foundation underlying for our approach. We model brain activity as signals supported on a graph $G = (V, E)$, where each node corresponds to a brain region and edges represent structural and functional connectivity. To capture structured and heterogeneous interactions between the regions, we represent these signals using a sheaf over a graph. We first introduce cellular sheaves and their algebraic representation, and then describe the associated sheaf Laplacian operator that generalizes classical graph diffusion.

A cellular sheaf $\mathcal{F}$ on a undirected graph $G = (V, E)$ assigns vector spaces to nodes and edges called *stalks*, together with linear maps that relate them called *restriction maps*. Formally, it consists of:

- A vector space $\mathcal{F}(i) \cong \mathbb{R}^{d_i}$ for each node $i \in V$
- A vector space $\mathcal{F}(e_{ij}) \cong \mathbb{R}^{d_e}$ for each edge $e_{ij} \in E$
- Restriction maps $\rho_{i \rightarrow e_{ij}} : \mathcal{F}(i) \rightarrow \mathcal{F}(e_{ij})$, $\rho_{j \rightarrow e_{ij}} : \mathcal{F}(j) \rightarrow \mathcal{F}(e_{ij})$ for each edge $e_{ij} \in E$.

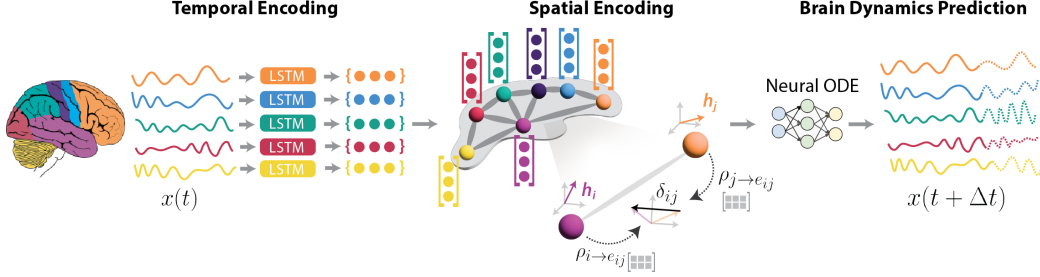


Figure 1: BrainDyn encodes each region’s temporal history with an LSTM, and then uses learnable sheaf restriction maps to align and modulate interactions across brain regions. The resulting spatially encoded latent state is evolved with a neural ODE to predict future brain dynamics.

We denote node features as $x_i \in \mathcal{F}(i)$. For each edge e_{ij} , the restriction maps project node features into a shared edge space $\mathcal{F}(e_{ij})$. This defines how information from neighboring nodes is aligned before interaction. In contrast to standard graph neural networks, where node features are directly aggregated, the sheaf structure allows features to be transformed differently along each edge prior to aggregation.

To model diffusion of signals over a sheaf, we define the associated *sheaf Laplacian* operator. The sheaf Laplacian is a symmetric, positive semi-definite block matrix. For each edge e_{ij} , we fix an arbitrary orientation $i \leftarrow j$ and observe the restriction maps $\rho_{i \rightarrow e_{ij}}$ and $\rho_{j \rightarrow e_{ij}}$ project node features into a shared edge space $\mathcal{F}(e_{ij})$, and the differences in their projects measures the inconsistency between neighboring nodes after alignment.

Specifically, we define the space of 0-cochains, $C^0(G, \mathcal{F})$, to be the direct sum of all the $\mathcal{F}(i), i \in V$ and similarly define the set space of 1-cochains, $C^1(G, \mathcal{F})$, as the direct sum of all the $\mathcal{F}(e_{ij}), e_{ij} \in E$. We then define $\delta : C^0(G, \mathcal{F}) \rightarrow C^1(G, \mathcal{F})$ by

$$\delta(X)(e_{ij}) = \rho_{i \rightarrow e_{ij}} x_i - \rho_{j \rightarrow e_{ij}} x_j \quad (1)$$

for $X \in C^0(G, \mathcal{F})$. The sheaf Laplacian is then defined as $L_{\mathcal{F}} = \delta^{\top} \delta$ where $\delta^{\top} : C^1(G, \mathcal{F}) \rightarrow C^0(G, \mathcal{F})$ is the adjoint of δ . The matrix $L_{\mathcal{F}}$ has block structure with its diagonal blocks given by $\mathcal{L}_F(i, i) = \sum_{e_{ij} \in E} \rho_{i \rightarrow e_{ij}}^{\top} \rho_{i \rightarrow e_{ij}} x_i$, and its off diagonal blocks are given by $\mathcal{L}_F(i, j) = -\rho_{i \rightarrow e_{ij}}^{\top} \rho_{j \rightarrow e_{ij}} x_i$ for $e_{ij} \in E$ and $\mathcal{L}_F(i, j) = 0$ otherwise [47]. This allows us to write

$$(L_{\mathcal{F}} X)_i = \sum_{\{j: e_{ij} \in E\}} \rho_{i \rightarrow e_{ij}}^{\top} (\rho_{i \rightarrow e_{ij}} x_i - \rho_{j \rightarrow e_{ij}} x_j) = \sum_{\{j: e_{ij} \in E\}} \rho_{i \rightarrow e_{ij}}^{\top} \delta(X)(e_{ij}). \quad (2)$$

Therefore the sheaf Laplacian can be seen as aggregating these discrepancies from edges back to the nodes. It generalizes the classical graph Laplacian by incorporating edge-dependent transformations before computing differences. Intuitively, it measures how inconsistent node features are across the graph after being aligned with the restriction maps. When neighboring nodes are consistent under these transformations, the Laplacian vanishes, while discrepancies induce diffusion that drives the system towards agreement.

4 BrainDyn Methodology: A Neural Sheaf ODE Network

We now describe the proposed method, **BrainDyn**, a sheaf-based neural dynamical model for learning continuous-time evolution signals on a graph $G = (V, E)$. Each node $i \in V$ corresponds to a brain region, we observe time-series signals $x_i(t) \in \mathbb{R}$. Our key idea is to represent node history in stalks, and expressive edge connections via sheaf restriction maps.

Our framework consists of three components:

1. **Memory-based node stalks:** Neural activity is inherently history-dependent, with the current state of a region reflecting memory of accumulated past dynamics. To capture this behavior, we encode the recent temporal history of each node using an LSTM, into each node stalk, enabling the model to learn intrinsic temporal structure within each region.

2. **Edge modulation via sheaf restriction maps:** Communications between brain regions is structured and heterogeneous, with connections transforming and modulating signals in distinct ways. We model this using a learnable restriction map that aligns node representations, together with feature-wise gating that enables channel-specific and direction-dependent coupling.
3. **Continuous-time evolution using a neural ODE:** Neural activity evolves continuously over time, driven by both intrinsic dynamics and interactions across regions. We model this using a neural ODE whose vector field depends on interactions between nodes, enabling the generation of dynamics based on neural information flow.

In brain networks, each region’s activity depends on both its own past activity and inputs from the connected regions. We therefore encode temporal history at each node, model inter-region interactions through sheaf-based modulation operators that act on these representations, and evolve the resulting features using a neural ODE. This yields a continuous-time model of neural dynamics in which both local temporal structure and connection-specific interactions jointly determine future activity.

4.1 Prior Graph Construction

We construct an initial prior graph, $\mathcal{P}$, using Granger causality computed only from the observed context window provided as input to the model. For each training example, Granger causality is estimated over the input time-lapse sequence, and connections with sufficiently strong scores are added as edges in $\mathcal{P}$. To control graph sparsity, we retain the top- k strongest Granger causal interactions for each node, where k is selected based on the expected average connectivity of the neural system. The forecast window is not used when constructing $\mathcal{P}$, ensuring that no future information is introduced during prediction. Since Granger causality captures directed temporal dependencies, the resulting graph reflects putative information flow between regions. This graph is then used to initialize the sheaf structure and regularize the learned interactions during training.

4.2 Memory encoding of node stalks

Neural signals exhibit complex temporal structure, where past activity strongly influences future dynamics. To capture this structure, we represent each node as an embedding that summarizes the temporal history. Given a temporal window of length $T = p - q + 1$ for node i , we consider a sequence $[x_i(t_p), x_i(t_p + 1) \dots, x_i(t_q)]$ and encode it using an LSTM:

$$h_i(t) = \text{LSTM}(x_i(t_p : t_q)) \quad (3)$$

where $h_i(t) \in \mathbb{R}^d$ corresponds to the hidden state of the sequence. This formulation allows us to compress complex temporal patterns such as oscillations, delays, and long-range dependencies into a representation that can be interpreted as a latent dynamical state of a region, encoding its recent activity trajectory. We interpret $h_i(t)$ as the stalk representation associated with node i . We define the global latent state by stacking all node stalks in Eqn (4).

$$H(t) = [h_1^\top(t), h_2^\top(t), \dots, h_N^\top(t)]^\top \in \mathbb{R}^{N \times d}. \quad (4)$$

4.3 Learnable Sheaf Laplacians via Learned Restriction maps with Attention Weights

Interactions between brain regions are heterogeneous and context-dependent, with different connections modulating signals in different ways. To model this, we construct a learnable version of the Sheaf Laplacian introduced in Section 3. For each edge e_{ij} , we let $\rho_{i \rightarrow e_{ij}}, \rho_{j \rightarrow e_{ij}} \in \mathbb{R}^{d \times d}$ be learnable restriction maps which define how node features are transported into a shared edge space and reparameterized when transmitted across a connection, thereby allowing different edges to encode different modes of interaction $h_{i \rightarrow e_{ij}} = \rho_{i \rightarrow e_{ij}} h_i$, $h_{j \rightarrow e_{ij}} = \rho_{j \rightarrow e_{ij}} h_j$, where h_i and h_h are as in Eqn. (4).

The aligned representations do not contribute equally to neural communication. In neural systems, different components of activity such as distinct dynamical modes may be selectively emphasized or suppressed depending on context. To capture this behavior, we introduce learnable attention coefficients that modulate the contribution of neighboring nodes during interaction. Specifically for each edge e_{ij} , we compute scalar attention coefficients from the transformed node representations using learnable weight vectors:

$$\alpha_i = \sigma(a^\top h_{i \rightarrow e_{ij}}), \quad \alpha_j = \sigma(a^\top h_{j \rightarrow e_{ij}}), \quad (5)$$

where $a \in \mathbb{R}^d$ is a learnable parameter vector shared across edges and $\sigma(\cdot)$ denotes the sigmoid activation function. These coefficients determine how strongly each transformed node representation contributes to the interaction along the edge. We then, analogous to Eqn. (1), compute an edge-level discrepancy between neighboring regions:

$$\delta_{ij} = \alpha_i h_{i \rightarrow e_{ij}} - \alpha_j h_{j \rightarrow e_{ij}}. \quad (6)$$

Finally, analogous to Eqn. (2), we may define the learnable sheaf Laplacian by

$$(L_{\mathcal{F}} H)_i = \sum_{\{j: e_{ij} \in E\}} \rho_{i \rightarrow e_{ij}}^{\top} (\alpha_i h_{i \rightarrow e_{ij}} - \alpha_j h_{j \rightarrow e_{ij}}) = \sum_{\{j: e_{ij} \in E\}} \rho_{i \rightarrow e_{ij}}^{\top} \delta_{ij}. \quad (7)$$

As in Section 3, globally, the sheaf Laplacian corresponds to the operator $L_{\mathcal{F}} = \delta^{\top} \delta$, where δ denotes the coboundary operator. Intuitively, it measures disagreement between neighboring regions after their representations have been aligned in edge-specific shared spaces. Regions that are well aligned produce small discrepancy signals, while mismatches lead to larger responses.

This reflects how neural communication is driven by differences in activity across regions. The learnable gating weights allow the model to selectively modulate which components of the signal are transmitted across connections, while the restriction maps determine how those signals are geometrically aligned. Together, these mechanisms yield a highly expressive interaction framework that captures how structured connectivity and channel-specific modulation jointly shape communication between brain regions.

4.4 Sheaf Message Passing

Analogous to graph neural networks, we perform iterative message passing over the sheaf to diffuse information between neighboring brain regions while respecting the geometry induced by the restriction maps. Let $H^{(0)} = H(t)$ denote the stacked node stalk representations from Eqn. 4. We then perform L rounds of sheaf message passing according to

$$H^{(l)} = (I - L_{\mathcal{F}}) H^{(l-1)}, \quad l = 1, \dots, L, \quad (8)$$

where I denotes the identity operator and $L_{\mathcal{F}}$ is the learnable sheaf Laplacian defined in Eqn. 7. This operator performs diffusion over the sheaf by propagating discrepancy signals between neighboring nodes after alignment in shared edge-specific spaces. Repeated application enables each node to aggregate information from progressively larger graph neighborhoods, analogous to receptive field expansion in graph neural networks. Unlike standard graph message passing, however, the sheaf formulation allows each edge to apply a distinct learnable transformation prior to communication, enabling heterogeneous and direction-sensitive interactions between brain regions.

The resulting representation after L rounds of message passing is denoted by $H^{(L)}$, with $h_i^{(L)}$ corresponding to the final representation of node i .

4.5 Continuous-time dynamics using a neural ODE

Neural activity evolves continuously over time, driven by both intrinsic dynamics within each brain region and interactions with other regions. These interactions are not static, they depend on the current state of the system and reflect ongoing coordination between regions. In order to capture this behavior, we model the evolution of neural signals using a continuous-time dynamical system whose vector field is defined through the sheaf-based interaction operator.

Formally, we let the evolution of each node be given by

$$\frac{dx}{dt} = f_{\theta}(h_i^{(L)}) = \text{MLP}_{\theta}(h_i^{(L)}). \quad (9)$$

This defines a dynamical system in which changes in neural activity are driven by discrepancies between regions, as captured by the sheaf Laplacian. In other words, the model evolves the system toward configurations that are consistent with the learned sheaf structure. Since the interaction operator incorporates both edge-specific transformations and attention weights, the resulting dynamics are highly expressive. This formulation reflects the principle that neural communication is driven by differences in activity across regions. The brain dynamically adjusts interactions to reduce

mismatches, leading to coordinated patterns such as synchronization and desynchronization across networks. The system is then solved using a numerical ODE solver:

$$\hat{x}_i(t) = \text{ODESolve}(f_\theta, x_i(t_q), [t_q, t_r]) \quad (10)$$

which produces continuous-time trajectories of neural activity.

To train the model, we optimize the reconstruction error between the predicted and observed neural trajectories using a mean squared error objective. In addition, we introduce two regularization terms. First, we encourage sparse interactions between regions by penalizing the magnitude of the edge discrepancy signals:

$$\mathcal{L}_{\text{sparse}} = \sum_{e_{ij} \in E} \|\delta_{ij}\|_1, \quad (11)$$

where δ_{ij} is defined in Eqn. 6. This encourages the model to learn parsimonious communication patterns between brain regions. Second, we encourage the learned interactions to remain close to the prior graph, $\mathcal{P}$, via:

$$\mathcal{L}_{\text{prior}} = \sum_{e_{ij} \in E \cup E_{\mathcal{P}}} \left| \|\delta_{ij}\|_2 - \mathbf{1}_{\{e_{ij} \in E_{\mathcal{P}}\}} \right|, \quad (12)$$

where $\mathbf{1}_{\{e_{ij} \in E_{\mathcal{P}}\}}$ is an indicator function that evaluates to 1 if edge e_{ij} belongs to the prior graph $\mathcal{P}$ and 0 otherwise. This regularization encourages edges present in the prior graph to exhibit stronger interaction discrepancies while suppressing unsupported interactions outside the prior structure.

The overall training objective combines the trajectory reconstruction loss with the sparsity and prior graph regularization terms:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_1 \mathcal{L}_{\text{sparse}} + \lambda_2 \mathcal{L}_{\text{prior}}, \quad (13)$$

where $\mathcal{L}_{\text{MSE}}$ denotes the mean squared error between the predicted and observed neural trajectories, while λ_1 and λ_2 control the contribution of the sparsity and prior graph regularization terms, respectively. Together, these objectives encourage the model to accurately forecast neural dynamics while learning sparse interaction patterns that remain consistent with known anatomical or functional connectivity priors. The computational complexity of the method is provided in Appendix D.

5 Empirical Results

We validate BrainDyn through three complementary evaluations designed to test whether it captures the structure of neural dynamics rather than merely fitting observed time series. First, in Section 5.1 we evaluate forecasting performance on resting-state fMRI from the Philadelphia

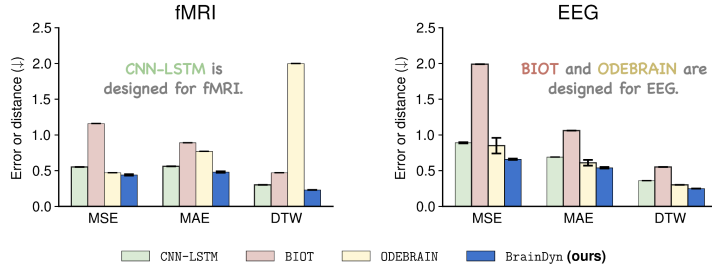


Figure 2: BrainDyn outperforms modality-specialized models.

Neurodevelopmental Cohort (PNC) [33] and scalp EEG from the Temple University Hospital Seizure Corpus (TUSZ) [34], covering both slow hemodynamic fluctuations and fast electrophysiological activity across healthy and clinical populations. Notably, most existing methods for brain dynamics modeling are tailored to either fMRI or EEG, and tend to perform much better within their target modality. In contrast, BrainDyn performs strongly across both modalities and outperforms the strongest modality-specific baselines in each setting (Figure 2). Second, in Section 5.2, we use synthetic neuronal activity generated with the NEST simulator [35] to assess whether a model trained only on unperturbed activity can extrapolate to perturbed, out-of-distribution dynamics. Experimental setup details are provided in Appendix C.

⁰Code Available at: <https://github.com/KrishnaswamyLab/BrainDyn>

Methods for comparison. Across these settings, we compare BrainDyn with representative recurrent (CNN-LSTM [41, 42]), transformer (BIOT [43]), dynamic graph (EvolveGCN [44]), and graph neural ODE baselines (ODEBRAIN [45], RiTINI [46]). We also report two ablations: *BrainDyn (Graph)*, which removes the learnable restriction maps and reduces the sheaf structure to a standard graph, and *BrainDyn (w/o LSTM)*, which removes the temporal encoder and uses raw windowed inputs as node stalks. Datasets and preprocessing are described in Appendix B, whereas baseline implementations and hyperparameters are in Appendix E.

5.1 Brain Dynamics Forecasting

We first assess BrainDyn’s ability to forecast neural dynamics across two modalities with very different temporal and spatial characteristics. The PNC resting-state fMRI dataset [33] comprises 1188 subjects scanned at rest, parcellated into 400 cortical regions using the Schaefer atlas [48], and sampled at a TR of 3 seconds. The TUSZ EEG dataset [34] contains continuous scalp recordings from 315 patients, with 19 channels resampled to 200 Hz; following the protocol described in Appendix B, we use a binary subset of seizure and matched non-seizure windows. For both datasets, we train each model to predict a forecast window of length 30 timepoints (90 seconds of fMRI recording, 0.05 seconds of EEG recording) given a context window of length 10 timepoints (30 seconds of fMRI recording, 0.15 seconds of EEG) recording, and evaluate on held-out subjects using 5-fold cross-validation.

We report mean squared error (MSE) and mean absolute error (MAE) as standard pointwise metrics, together with a normalized dynamic time warping (DTW) distance that measures shape-level agreement while remaining tolerant of small temporal offsets (see Appendix F for a discussion of why correlation-based metrics are inadequate in this setting).

Table 1 summarizes the results. On resting-state fMRI, BrainDyn achieves the lowest error across all three metrics, reducing MSE by 22% relative to the strongest baseline (ODEBRAIN, 0.66 vs. 0.85) and DTW distance by 17% (0.25 vs. 0.30). Results on TUSZ EEG shows the same pattern: BrainDyn again outperforms all baselines, with clear improvement over the best baselines.

For both fMRI and EEG, the performance advantages against non-graph baselines (CNN-LSTM, BIOT) and dynamic graph baselines that lack continuous-time integration (EvolveGCN) are substantially larger, suggesting that the combination of structured connectivity and continuous-time evolution make significant contribution to the superior brain dynamics modeling. Equally informative are the ablations: removing the LSTM encoder (*BrainDyn w/o LSTM*) degrades MSE (0.66 to 1.60 on fMRI, and 0.44 to 1.00 on EEG), whereas removing the sheaf structure (*BrainDyn Graph*) also yields a substantial drop, confirming that both components contribute and that neither alone is sufficient.

5.2 Forecasting Perturbed Dynamics

A virtual brain model is most useful when it can generalize to interventions that were not observed during training. To test this capability, we used the NEST simulator [35] to generate activity from networks of 100 integrate-and-fire neurons with alpha-shaped postsynaptic currents. We generated 10,000 independent instantiations of this network, each with randomly sampled neuron positions

Table 1: Prediction of resting-state fMRI dynamics on the PNC dataset and EEG dynamics on the TUSZ dataset. Metrics reported as mean $\pm$ std across 5-fold cross-validation. Lower is better. Best results are bolded.

Method	fMRI [33]			EEG [34]		
	MSE $\downarrow$	MAE $\downarrow$	DTW $\downarrow$	MSE $\downarrow$	MAE $\downarrow$	DTW $\downarrow$
CNN-LSTM [41, 42]	0.89 $\pm$ 0.01	0.69 $\pm$ 0.00	0.36 $\pm$ 0.00	0.55 $\pm$ 0.00	0.56 $\pm$ 0.00	0.30 $\pm$ 0.00
BIOT [43]	1.99 $\pm$ 0.00	1.06 $\pm$ 0.00	0.55 $\pm$ 0.00	1.16 $\pm$ 0.00	0.89 $\pm$ 0.00	0.47 $\pm$ 0.00
EvolveGCN [44]	1.00 $\pm$ 0.04	0.71 $\pm$ 0.02	0.31 $\pm$ 0.00	0.65 $\pm$ 0.00	0.61 $\pm$ 0.00	0.29 $\pm$ 0.00
ODEBRAIN [45]	0.85 $\pm$ 0.11	0.61 $\pm$ 0.04	0.30 $\pm$ 0.00	0.47 $\pm$ 0.00	0.77 $\pm$ 0.00	0.27 $\pm$ 0.00
RiTINI [46]	1.91 $\pm$ 0.00	0.98 $\pm$ 0.00	0.47 $\pm$ 0.00	1.02 $\pm$ 0.00	0.81 $\pm$ 0.00	0.41 $\pm$ 0.00
BrainDyn (Graph)	1.26 $\pm$ 0.02	0.81 $\pm$ 0.02	0.40 $\pm$ 0.00	0.76 $\pm$ 0.01	0.68 $\pm$ 0.01	0.34 $\pm$ 0.00
BrainDyn (w/o LSTM)	1.60 $\pm$ 0.01	0.92 $\pm$ 0.01	0.43 $\pm$ 0.00	1.00 $\pm$ 0.02	0.80 $\pm$ 0.01	0.41 $\pm$ 0.00
BrainDyn	0.66 $\pm$ 0.01	0.54 $\pm$ 0.01	0.25 $\pm$ 0.00	0.44 $\pm$ 0.01	0.48 $\pm$ 0.01	0.23 $\pm$ 0.00

and subsequently connected according to a sparse directed small-world graph with 400 edges. All neurons in each network received the same shared Poisson background spike train at rate 1,000 Hz, and recurrent synaptic weights were uniform. We binned spike trains into 10 millisecond windows to obtain per-neuron firing-rate time series, which served as the observed signal for all models. To simulate perturbation, we silenced a single neuron over a contiguous window of random duration $D \in [240, 400]$ ms, beginning at a random onset time t_0 within the inner 80% of each 2,000-ms simulation. This intervention directly alters the activity of the targeted neuron and propagates through the network according to the simulated connectivity. Full simulation parameters and code are provided in Appendix B.

Table 2: Neural dynamics forecasting on synthetic NEST dataset of perturbed, out-of-distribution brain dynamics. Metrics reported as mean $\pm$ standard deviation across 5-fold cross-validation. Lower is better. Best results are bolded.

Method	MSE $\downarrow$	MAE $\downarrow$	DTW $\downarrow$
CNN-LSTM [41, 42]	0.896 ± 0.001	0.719 ± 0.001	0.378 ± 0.001
BIOT [43]	1.041 ± 0.015	0.790 ± 0.005	0.402 ± 0.003
EvolveGCN [44]	1.029 ± 0.008	0.785 ± 0.004	0.408 ± 0.004
ODEBRAIN [45]	0.702 ± 0.019	0.640 ± 0.008	0.293 ± 0.006
RiTINI [46]	0.904 ± 0.002	0.718 ± 0.001	0.352 ± 0.001
BrainDyn (Graph)	0.851 ± 0.001	0.715 ± 0.000	0.351 ± 0.000
BrainDyn (w/o LSTM)	0.748 ± 0.032	0.666 ± 0.014	0.309 ± 0.005
BrainDyn	0.671 ± 0.038	0.625 ± 0.016	0.289 ± 0.010

We constructed forecasting windows that straddle the perturbation onset, with 90% of the context before onset and 10% after onset. Thus, the input contains the perturbation onset and early network response, while the prediction window covers the subsequent perturbed trajectory. Critically, all models were trained only on unperturbed activity from the same networks and evaluated on perturbed windows without further training. This setting tests whether the learned dynamics extrapolate to out-of-distribution activity patterns, a key property for an *in silico* simulator of neural interventions.

Table 2 reports forecasting performance under this perturbation protocol. BrainDyn achieves the lowest MSE and DTW distance among all methods, with MSE = 0.671, MAE = 0.625 and DTW = 0.289. Compared with ODEBRAIN, the strongest baseline, BrainDyn reduces MSE by 4% (0.702 to 0.671), MAE by 2% (0.640 to 0.625), and DTW distance by 1% (0.293 to 0.289). The ablations further show that both components are important for intervention generalization: replacing the sheaf structure with a standard graph increases MSE to 0.851, and removing the LSTM temporal encoder increases MSE to 0.748. These results suggest that BrainDyn’s combination of learnable restriction maps and temporal encoding improves extrapolation to perturbed neural dynamics.

6 Conclusion

We introduced BrainDyn, a sheaf neural ODE framework for modeling continuous-time neural dynamics. To our knowledge this is the first time cellular sheafs have been combined with neural ODEs. By combining temporal encoding, learnable restriction maps, and ODE-based evolution, BrainDyn captures heterogeneous interactions across neural systems. Across NEST simulations and real-world fMRI and EEG data, BrainDyn performs strongly across modalities, and generalizes to perturbed out-of-distribution dynamics. These results demonstrate the value of combining sheaf-based interaction modeling with continuous-time dynamics to learn generalizable representations of brain activity.

Limitations. BrainDyn currently models resting dynamics, which are in some sense self-sustaining dynamics with little external sensory input. While it is possible to potentially generate stimulus-driven dynamics with strong visual or auditory components such as movie-watching or audio inputs using this type of model, it would have to be trained in a setting where several neuronal units are simultaneously perturbed, which the current model has not undertaken.

Broader Impacts. BrainDyn improves our ability to model and simulate neural dynamics across modalities such as EEG and fMRI. Such models could support the study of neurological disorders

by enabling analysis of abnormal transient activity patterns, forecasting future neural dynamics, and serving as *in silico* testbeds for perturbation and intervention studies. We do not foresee direct negative societal impacts from this work, as the proposed framework is designed for scientific modeling of neural activity rather than high-risk applications involving surveillance, decision-making, or generative media.

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Technical Appendices

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A Extended Related Works

Neural dynamics modeling. A foundational insight from systems neuroscience is that neural population activity, despite high dimensionality, is organized by low-dimensional dynamics: coordinated trajectories through state space that implement computation, drive behavior, and distinguish brain states [36–38]. The traditional approach to characterizing these dynamics has nonetheless relied on static or time-averaged representations: pairwise functional connectivity matrices, region-level activation maps, and spectral power summaries. While these descriptors have driven decades of progress, they fundamentally discard temporal structure, collapsing rich trajectories into single-point summaries. Sliding-window functional connectivity methods partially address this by tracking how pairwise correlations evolve over time, but remain tethered to pairwise statistics and are sensitive to window length and stationarity assumptions. More recent work has reframed brain signal modeling as a forecasting problem: given a history of multivariate activity, predict its near-future trajectory. This framing has proven productive in the EEG literature, where predicting future signal states motivates representation learning over the full spatiotemporal signal rather than hand-crafted features [39]. BrainDyn operates in this forecasting setting, treating dynamics prediction as the primary objective and downstream tasks as a means of evaluating learned representations.

Graph neural networks for brain signals. Graph neural networks provide a natural inductive bias for brain data, where electrodes, parcellated regions, or individual neurons can be modeled as nodes and anatomical or functional relationships as edges. Spectral and spatial GCN variants have been applied to fMRI and EEG alike, with diffusion-based message passing proving especially effective for seizure detection and classification [39]. Dynamic graph settings, where connectivity changes over time, have been addressed by architectures such as EvolveGCN [44], which allows graph filter weights to evolve as a function of temporal context, and temporal graph network variants that jointly model recurrent state and graph structure [45]. Despite these advances, standard message-passing architectures share a common assumption: all nodes communicate in the same feature space, with aggregation performed by scalar summation or averaging. In the brain, this assumption is questionable: distinct regions differ systematically in the dimensionality, geometry, and content of their representations. Repeated averaging across heterogeneous regions leads to over-smoothing, progressively homogenizing node states and erasing the regional specificity that is a defining characteristic of neural computation. These limitations motivate a richer interaction model.

Cellular sheaves and sheaf neural networks. A cellular sheaf over a graph assigns a vector space, termed a stalk, to each node and edge, together with linear restriction maps that relate neighboring

stalks. The sheaf Laplacian, derived from these maps, generalizes the classical graph Laplacian by measuring inconsistency between node features only after they have been transformed into a shared edge space [26]. Hansen and Gebhart [26] first proposed building neural networks around this structure, showing that sheaf convolutions generalize GCNs and naturally handle asymmetric and heterogeneous relational structure. Bodnar et al. [28] provided a comprehensive theoretical treatment of Neural Sheaf Diffusion, establishing that non-trivial sheaves can prevent over-smoothing and accommodate heterophilic graphs where standard GNNs degrade. Barbero et al. [27] further connected sheaf learning to Riemannian geometry, proposing connection Laplacians derived from local tangent space alignment as a geometrically principled approach to constructing restriction maps. In aggregate, this body of work establishes that equipping each edge with a learnable linear transformation yields strictly more expressive diffusion processes than scalar aggregation. BrainDyn applies this insight to neural dynamics: restriction maps are learned per connection, allowing distinct brain regions to maintain separate representational geometries while interacting through structured, edge-specific transformations.

Neural ODEs and continuous-time dynamics on graphs. Neural ordinary differential equations parameterize the derivative of a hidden state with a neural network and integrate it using an adaptive numerical solver, enabling continuous-time modeling without fixed-step discretization [30]. Rubanova et al. [31] extended this to latent variable models for irregularly-sampled time series, where observations arrive at non-uniform intervals that recurrent architectures handle poorly. Graph neural ODEs further combine continuous-time latent dynamics with graph-structured message passing, allowing the vector field to depend on relational context [32]. ODEBRAIN constructed a dual encoder and integrated them through a gated neural ODE field to forecast evolving EEG graph representations [45]. RiTINI coupled space-time graph attention with a graph neural ODE to infer time-varying regulatory interaction graphs, demonstrating the value of continuous-time propagation for network-level dynamical inference [46]. A shared limitation of these approaches is that all node representations live in a single, homogeneous latent space. BrainDyn departs from this by embedding the ODE vector field within a sheaf, so the interaction structure driving continuous-time evolution is itself geometrically heterogeneous and connection-specific.

Biosignal transformers and sequence models. Transformer-based architectures have emerged as strong general-purpose encoders for biosignals. BIOT [43] tokenizes multichannel waveforms via short-time Fourier transforms and applies a Linformer over the resulting token sequence, enabling cross-dataset learning across heterogeneous EEG recording configurations. Related work on pretraining models has shown that self-supervised objectives over raw or frequency-domain representations yield representations that transfer to downstream clinical tasks. At a simpler level, hybrid convolutional-recurrent architectures capture local temporal features with CNNs before aggregating them with an LSTM, offering a compact baseline for multivariate forecasting [41, 42]. A common limitation of these sequence models is that they are largely agnostic to brain structure: channels or regions are treated as independent or fully-connected, without explicit modeling of anatomical or functional connectivity. This weakens their utility as *in silico* simulators, where the organization of the brain graph should constrain how predictions propagate across the network.

Generative and foundation models for neural data. The ambition of building a generative model of the brain has a long history in computational neuroscience. The Virtual Brain project helped establish the idea that empirical connectivity can be combined with dynamical systems to simulate whole-brain activity, motivating *in silico* brain models that can generate and forecast neural dynamics [4]. More recently, large-scale foundation models have brought a complementary data-driven approach, using broad neural and imaging datasets to learn representations that transfer across subjects, tasks, and clinical conditions. However, many of these approaches are still trained on largely static objectives: reconstruction, masked prediction, or contrastive alignment. This approach often does not explicitly model the continuous-time process that generates observed neural activity. BrainDyn takes a complementary approach: rather than predicting masked tokens or reconstructing static observations, it learns by forecasting the continuous trajectory of neural activity over a structured brain graph, grounding representation learning in the dynamics themselves. BrainDyn is a structure-aware, dynamics grounded pretraining framework for an *in silico* generative foundation model of the brain, analogous in spirit to virtual-cell models in biology.

B Datasets and Preprocessing

Resting-state fMRI (PNC). Resting-state BOLD was acquired under the Philadelphia Neurodevelopmental Cohort (PNC) protocol (single-site imaging; resting-state EPI with $TR = 3$ s in the cohort neuroimaging description). For this work, publicly released derivatives from the Reproducible Brain Charts (RBC) initiative were used: functional scans were processed with the Connectome Pipeline for the Analysis of Connectomes (C-PAC; `cpac_RBCv0`), including standard volume realignment and nuisance regression as distributed in that release. Activity was summarized using a fixed atlas-based parcellation, producing one mean BOLD time series per region per run. The atlas we used was the Schaefer 2018 cortical atlas (400 regions, 17-network solution) [48] in MNI152NLin6ASym space. Nuisance effects were addressed with a 36-parameter regression strategy, as documented for the RBC CPAC outputs used here. Each scan is thus represented as a multivariate time series of dimension 400 over T retained volumes after processing. The fMRI dataloader constructs windows over parcellated ROI time series with shape (400, 40): 400 ROI signals and 40 consecutive timepoints (30 context, 10 horizon). Using a stride between windows of 40, this yields 1,664 training windows and 356 windows each for validation and test (2,376 total windows). Before modeling, we apply per-window, per-ROI z-score normalization along the time axis.

Scalp EEG with focal epilepsy (TUSZ, binary subset). The Temple University Hospital EEG Seizure Corpus (TUSZ) provides continuous clinical scalp EEG stored in European Data Format (EDF) together with expert seizure annotations from patients evaluated in a comprehensive epilepsy surgery program. Rather than using the full heterogeneous event mix of the corpus, experiments use a binary subset of short temporal windows labeled as non-seizure (*clean*) versus seizure (*ictal*), constructed by subsampling background intervals and matched seizure segments so that evaluation is not dominated by long stretches of interictal activity or rare transition geometries. Annotation in TUSZ is defined on a standard 19-channel 10-20 montage. The EEG dataloader constructs windows of shape (19,40): 19 EEG channels and 40 consecutive time points (30 context, 10 horizon). Using a stride between windows of 40, this yields 10,000 training windows and 2,500 windows each for validation and test (15,000 total windows). Before modeling, we apply per-window, per-channel z-score normalization along the time axis.

NEST simulations. We generated synthetic neural activity using NEST [35] from directed small-world networks of 100 `iaf_psc_alpha` neurons. Each simulation lasted 2,000 ms with 0.1 ms resolution. Recurrent edges followed a directed small-world graph with `small_world_k= 8` and rewiring probability `small_world_beta= 0.1`, yielding 400 directed edges. All neurons received a shared Poisson background drive at 1,000 Hz, with uniform recurrent synaptic weight and delay. Spike trains were binned into 10 ms windows, converted to firing rates, and smoothed with a Gaussian kernel of standard deviation 20 ms. For perturbation experiments, we selected one neuron uniformly at random and silenced it over a contiguous intervention window sampled within the central portion of the trajectory, allowing the perturbation to propagate through the recurrent simulated graph. We saved the unperturbed and perturbed firing-rate trajectories, ground-truth adjacency matrices, graph seeds, bin edges, and perturbation metadata for 10,000 independently sampled network instantiations.

C Experimental Setup

BrainDyn is trained using the AdamW [49] optimizer with an initial learning rate of 10^{-3} and weight decay 10^{-5} . The learning rate is scheduled using ReduceLROnPlateau with a multiplicative decay of 0.5, a patience of 3 epochs, and minimum learning rate of 10^{-6} . The ODE is integrated using a fixed-step fourth-order Runge-Kutta (RK4) solver with step size $\Delta t = 1$, implemented via `torchdiffeq` [50]. Models are trained for up to 100 epochs with a batch size of 64, and the checkpoint with the lowest validation loss is retained. We perform 5-fold cross validation with an 80/10/10 train/validation/test split at the subject level. All experiments are run with automatic mixed precision (fp16) on a single NVIDIA H200 GPU.

D Computational Complexity

Per-sample computational cost of BrainDyn decomposes into three stages. The temporal encoder applies an L -layer LSTM to each node’s context window of length T , giving $O(NT(FD + LD^2))$ where F is the signal dimension and D is the hidden dimension. The sheaf Laplacian then operates on the encoded node representations: for each edge (u, v) it applies learned restriction maps $\rho \in \mathbb{R}^{D \times M}$ to project into the shared edge space, computes the inter-node disagreement, and pulls back to node space, costing $O(EDM)$ where $E = |\mathcal{E}|$ is the number of edges and M is the map dimension. Finally, the neural vector field which is a two layer MLP of width V is evaluated at each of $4S$ calls required by the RK4 integrator over a horizon of S steps, contributing $O(SN(DV + V^2))$. The overall per-sample complexity is therefore $O(NTD^2 + EDM + SNV^2)$, linear in forecast horizon S and graph size N for sparse graphs, with sheaf diffusion dominating for larger graphs.

E Baseline methods

CNN-LSTM. The common hybrid design in multivariate forecasting is convolutional feature extraction over time followed by a recurrent readout [41, 42]. We followed that design and implemented a per-node CNN-LSTM: each node’s context trace passes through a shallow temporal convolutional stack and an LSTM, and future steps are produced with learned step embeddings. This baseline captures local temporal structure but ignores brain connectivity.

EvolveGCN. EvolveGCN [44] maintains node states on the graph while allowing the GCN weights to evolve as a function of the temporal context, yielding a standard dynamic-graph baseline that uses connectivity without a continuous-time latent integrator.

BIOT. We use the public BIOT encoder [43], a transformer over short-time Fourier tokens of multichannel waveforms, with a lightweight forecasting head on the pooled embedding. On EEG the encoder sees the multichannel context window directly; on parcellated fMRI we expand each TR into a constant pseudo-waveform block so the spectrogram front end receives a compatible time axis.

ODEBRAIN. We compare to the ODEBRAIN [45] framework [45], which fuses spectral graph features with raw temporal information to form an initial state and integrates a learned neural ODE field to forecast evolving graph-structured representations of EEG.

RiTINI. RiTINI [46] combines space-time graph attention with graph neural ODEs for dynamical systems on a graph prior. Our baseline preserves that decomposition (temporal encoding, graph attention on the FC edges, then ODE integration in latent space) but is reimplemented for batched (batch, time, node) tensors and our FC graph construction, isolating the role of sheaf-structured coupling relative to attention plus continuous-time propagation on the same topology.

F Trajectory metrics

Mean squared error and mean absolute error are the standard scalar summaries in time series forecasting. They measure the pointwise deviation between a predicted trajectory and the observed one at each time step, which makes them straightforward and easy to compare across methods. Relying on them alone, however, can be misleading when the goal is to evaluate whether a model captures the local dynamics of the signal rather than just its mean level. A forecast that is slightly shifted forward or backward in time can incur a large MSE or MAE even if it reproduces the shape, frequency content, and transient structure of the signal almost exactly. Conversely, a prediction that misses those features but stays close in absolute value at every step may score well.

Pearson and Spearman correlation are a natural candidate to complement MSE and MAE because they are scale-invariant and sensitive to relative movement. They are, however, not a robust solution for this purpose. Both coefficients are computed between values at matching time indices, so they mainly measure whether the two series move in the same direction at the same moments. A shared low-frequency component such as a drift or a slow baseline trend inflates correlation regardless of whether the forecast captures anything meaningful about the local signal structure. Figure S1 illustrates this failure mode: the observed trajectory contains oscillatory dynamics that are entirely

absent from the forecast, yet PCC and Spearman correlation both remain near 0.93 because the two series share the same linear trend.

Dynamic time warping offers a more appropriate comparison for this setting. Rather than aligning series at fixed time indices, DTW finds an optimal warping path between them that allows one series to be stretched or compressed locally to match the other, penalizing the cost of alignment along the way. This makes the resulting distance sensitive to differences in shape and local timing while being tolerant of small temporal offsets, which is precisely the kind of comparison that pointwise metrics and correlation miss. We report a normalized DTW distance throughout, defined as the total path cost divided by path length, so the score is comparable across trajectories of different lengths.

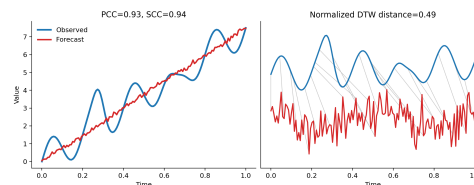


Figure S1: The same observed (blue) and forecast (red) trajectories shown in two views. Left: high correlation scores because both series share a linear trend. Right: Gray segments indicate the DTW warping path between the processed traces (show without trend for visualization) yielding a high normalized DTW distance (lower is better).